# VIII. Mini Project – Healthcare Analytics

## AIM:

To clean, process, analyze, and visualize healthcare data using Python libraries such as Pandas, Matplotlib, and Seaborn to identify patterns in patient admissions, medical conditions, treatment costs, and hospital departments.

## DOMAIN:

Healthcare Analytics

## PROBLEM STATEMENT:

Hospitals generate patient and treatment data across departments, admission dates, medical conditions, and costs. Analyzing these records manually makes it difficult to identify admission trends, compare department workloads, and understand cost distributions. A data-driven visualization framework is needed to summarize healthcare information and support clear exploratory analysis.

## CODE:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from google.colab import files

uploaded = files.upload()
df = pd.read_csv(next(iter(uploaded)))
df = df.drop_duplicates()
df['Admission_Date'] = pd.to_datetime(df['Admission_Date'])

sns.set_theme(style="whitegrid")

df['Department'].value_counts().plot(
   kind='bar', figsize=(7,4), title='Patients by Department')
plt.ylabel('Number of Patients')
plt.tight_layout()
plt.show()

plt.figure(figsize=(7,4))
plt.hist(df['Age'], bins=10, edgecolor='black')
```

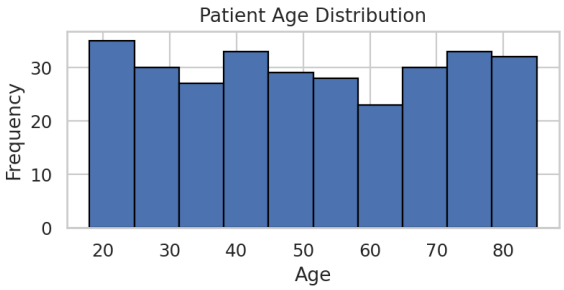
```
plt.title('Patient Age Distribution')
plt.xlabel('Age')
plt.ylabel('Frequency')
plt.tight_layout()
plt.show()

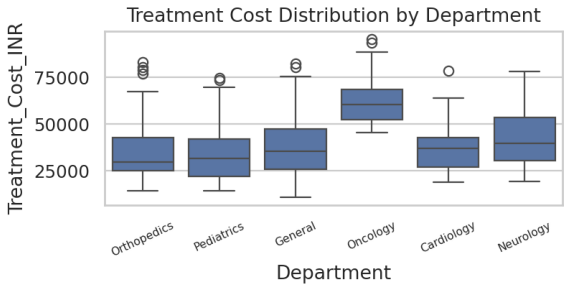
df['Medical_Condition'].value_counts().plot(
    kind='pie', autopct='%1.1f%%', figsize=(6,6))
plt.title('Distribution of Medical Conditions')
plt.ylabel('')
plt.show()
monthly = df.groupby(
    df['Admission_Date'].dt.to_period('M')).size()
monthly.index = monthly.index.astype(str)
monthly.plot(figsize=(9,4), marker='o', color='green')
plt.title('Monthly Hospital Admissions')
plt.xlabel('Month')
plt.ylabel('Admissions')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
plt.figure(figsize=(9,4))
sns.boxplot(data=df, x='Department', y='Treatment_Cost_INR')
plt.title('Treatment Cost Distribution by Department')
plt.xticks(rotation=30)
plt.tight_layout()
plt.show()
plt.figure(figsize=(7,4))
sns.heatmap(
    df[['Age','Length_of_Stay_Days',
        'Treatment_Cost_INR']].corr(),
    annot=True, cmap='coolwarm', fmt='.2f')
plt.title('Healthcare Data Correlation Heatmap')
plt.tight_layout()
plt.show()

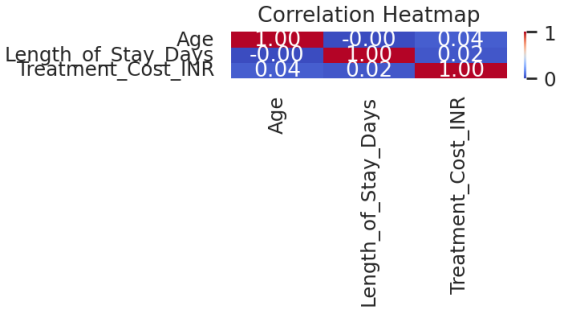
print("Python visualizations executed successfully!")
```

## OUTPUT:

## Python:

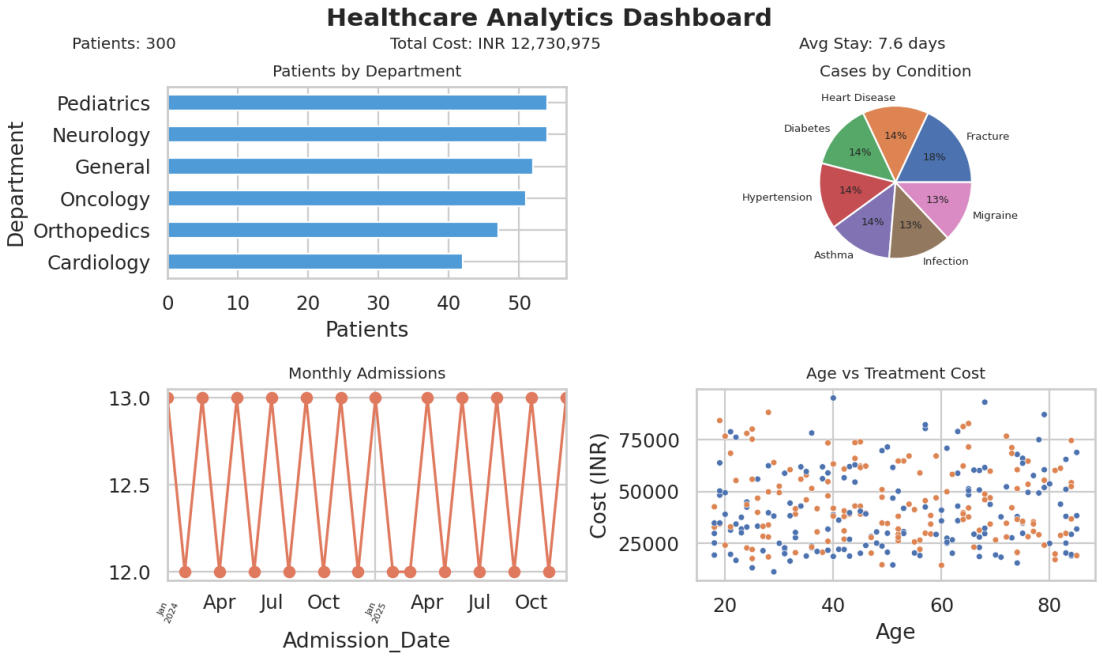


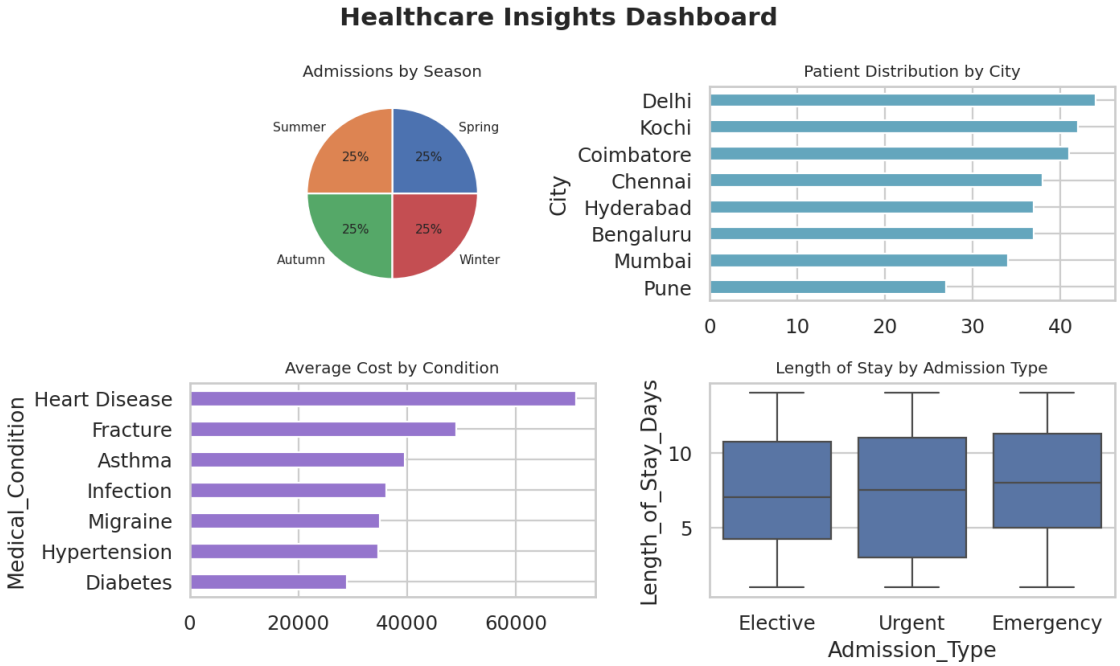

## Power BI-style dashboard (Python-generated mockup):

The following dashboard layout imitates common BI report elements for demonstration



## Tableau-style dashboard (Python-generated mockup):

This dashboard imitates a Tableau-style arrangement of charts.




## RESULT:

Thus, the healthcare dataset was cleaned and analyzed, and visualizations were generated successfully using Python to explore patient demographics, admission trends, department distribution, medical conditions, and treatment costs. The BI-style layouts are illustrative mockups.